usb_joystick_routed

Design Report

kicad-tools 0.24.0

Rev 1 | 2026-10-21 | jlcpcb-tier1

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	38 (37 SMD, 1 THT, 0 other)
Nets	27
Traces	910 segments
Vias	85
Board Size	80.0 x 60.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Cu	copper	0.035	–
dielectric 1	prepreg	0.2104	7628
In1.Cu	copper	0.0152	–
dielectric 2	core	1.065	FR4
In2.Cu	copper	0.0152	–
dielectric 3	prepreg	0.2104	7628
B.Cu	copper	0.035	–

Design Overview

Theory of Operation

USB joystick controller — ATmega32U4

USB bus powered, 8MHz crystal, AVR ISP programming

GPIO and ADC map is fixed to Microchip TQFP-44 pinout

Communication Interfaces

Protocol	Signals
SPI	ISP_MISO, ISP_MOSI, ISP_SCK
USB	USB_D+, USB_D-, VBUS

Assembly Notes

1 fine-pitch component

- **Fine-pitch components**: 1 (U1)

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: PASS

Schematic Overview

Schematic: usb_joystick

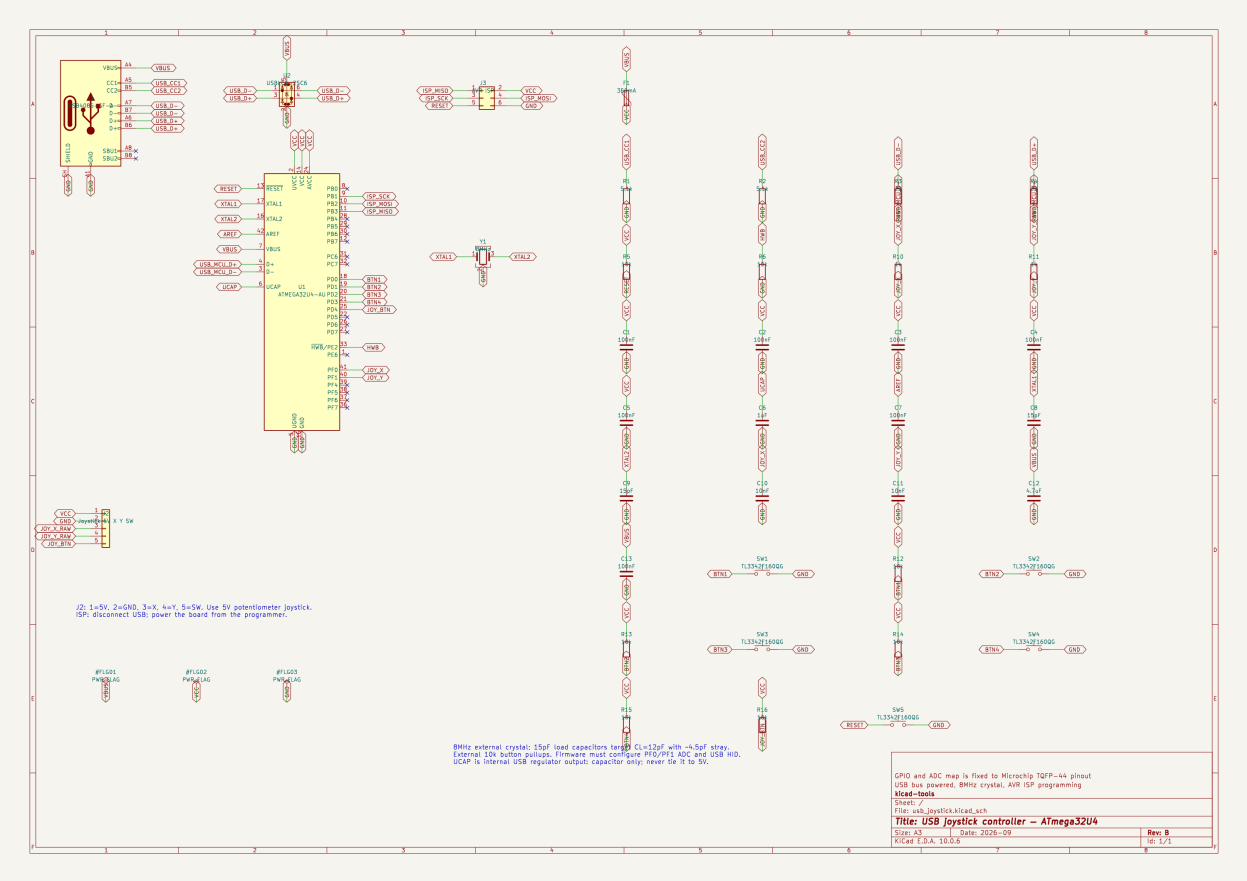


Figure 1: Schematic: usb_joystick

PCB Layout

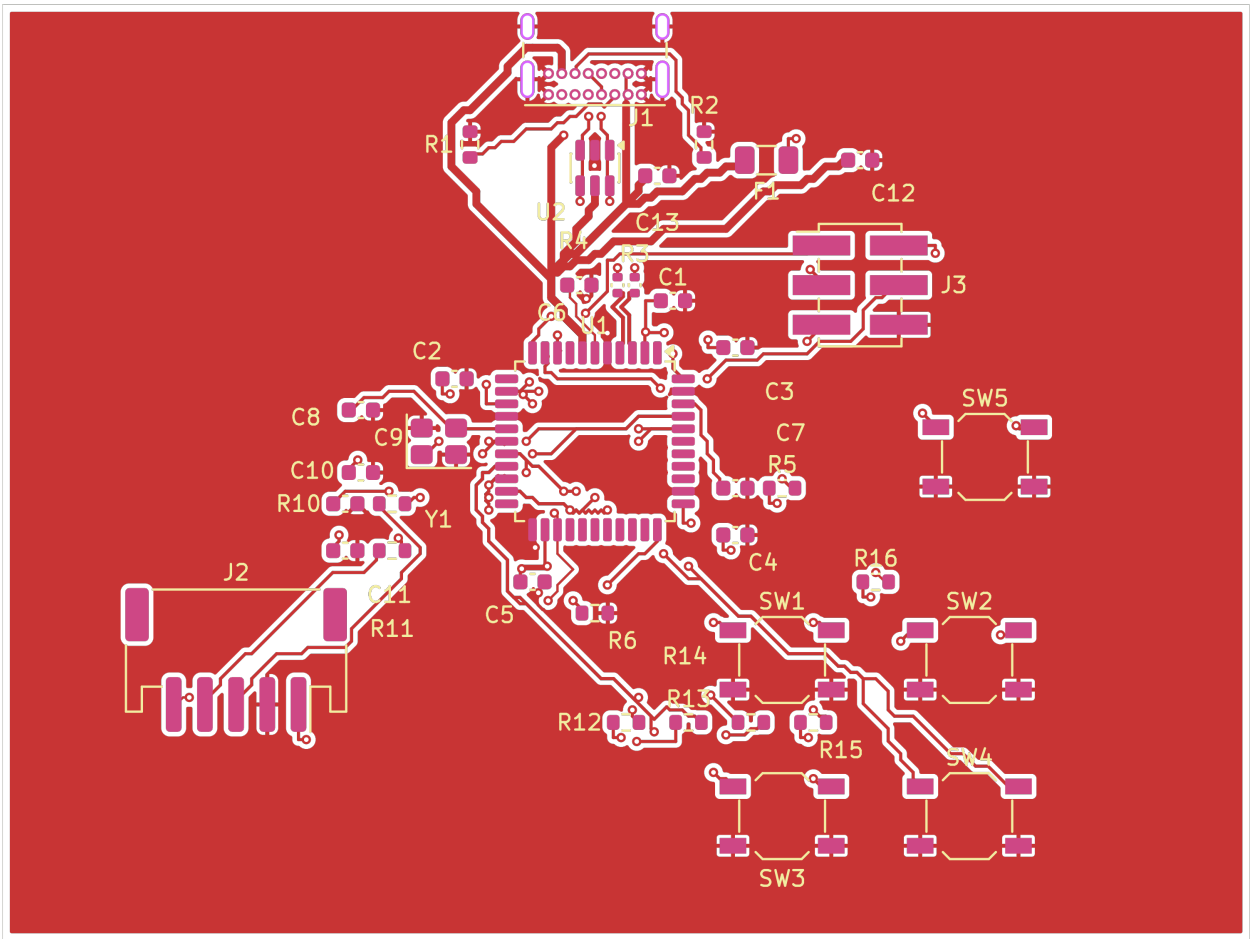


Figure 2: PCB Front

Copper

Assembly

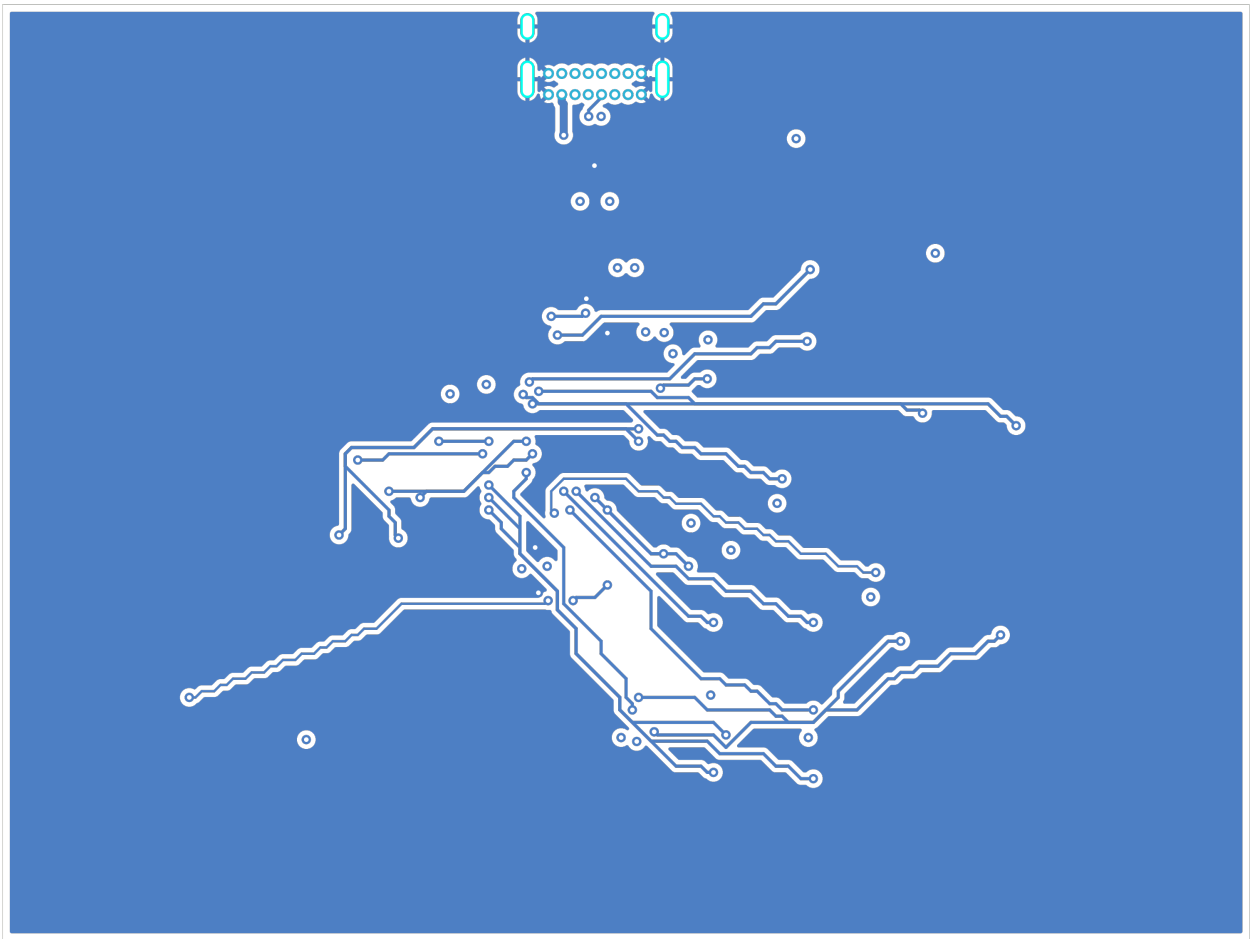


Figure 3: PCB Back

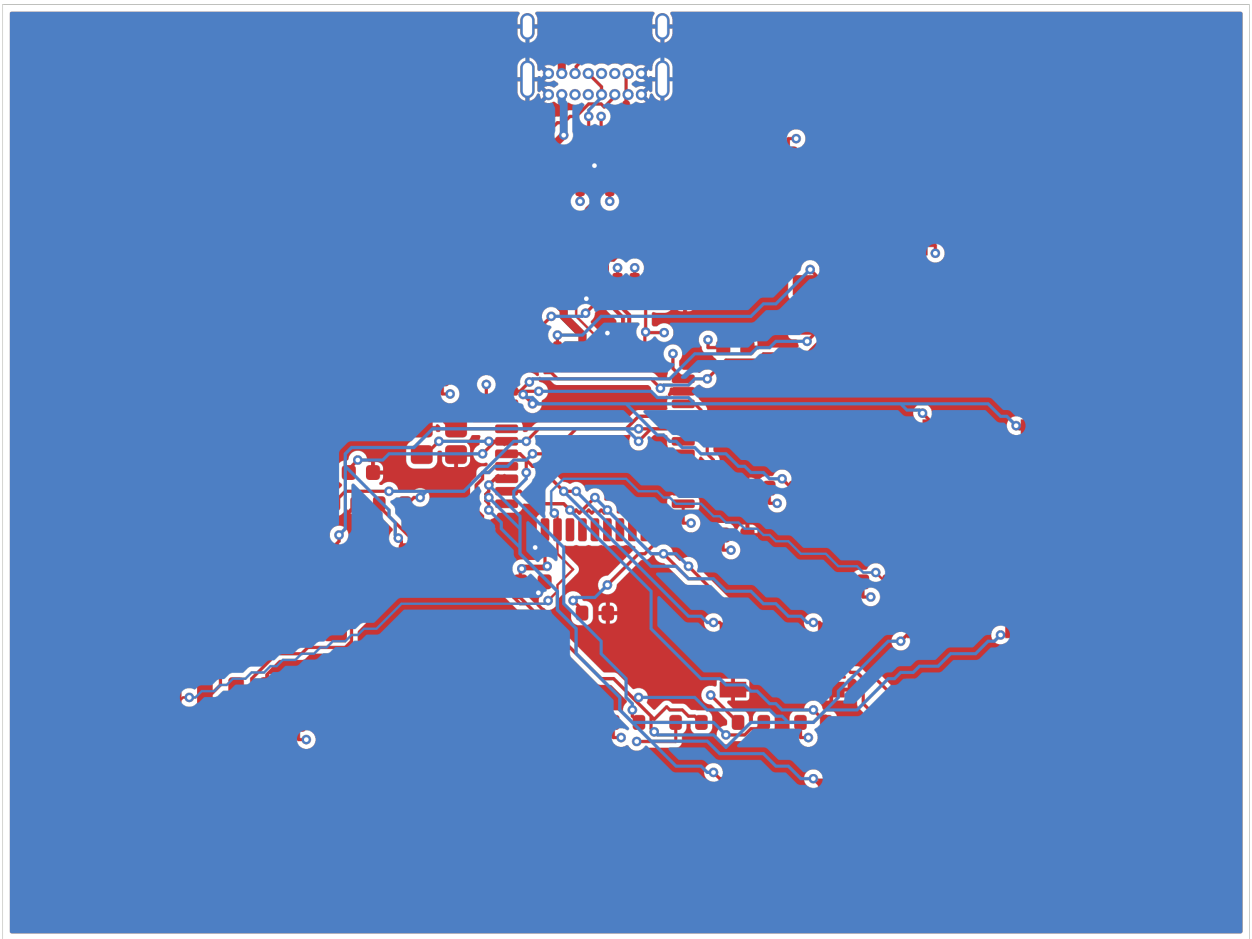


Figure 4: PCB Copper

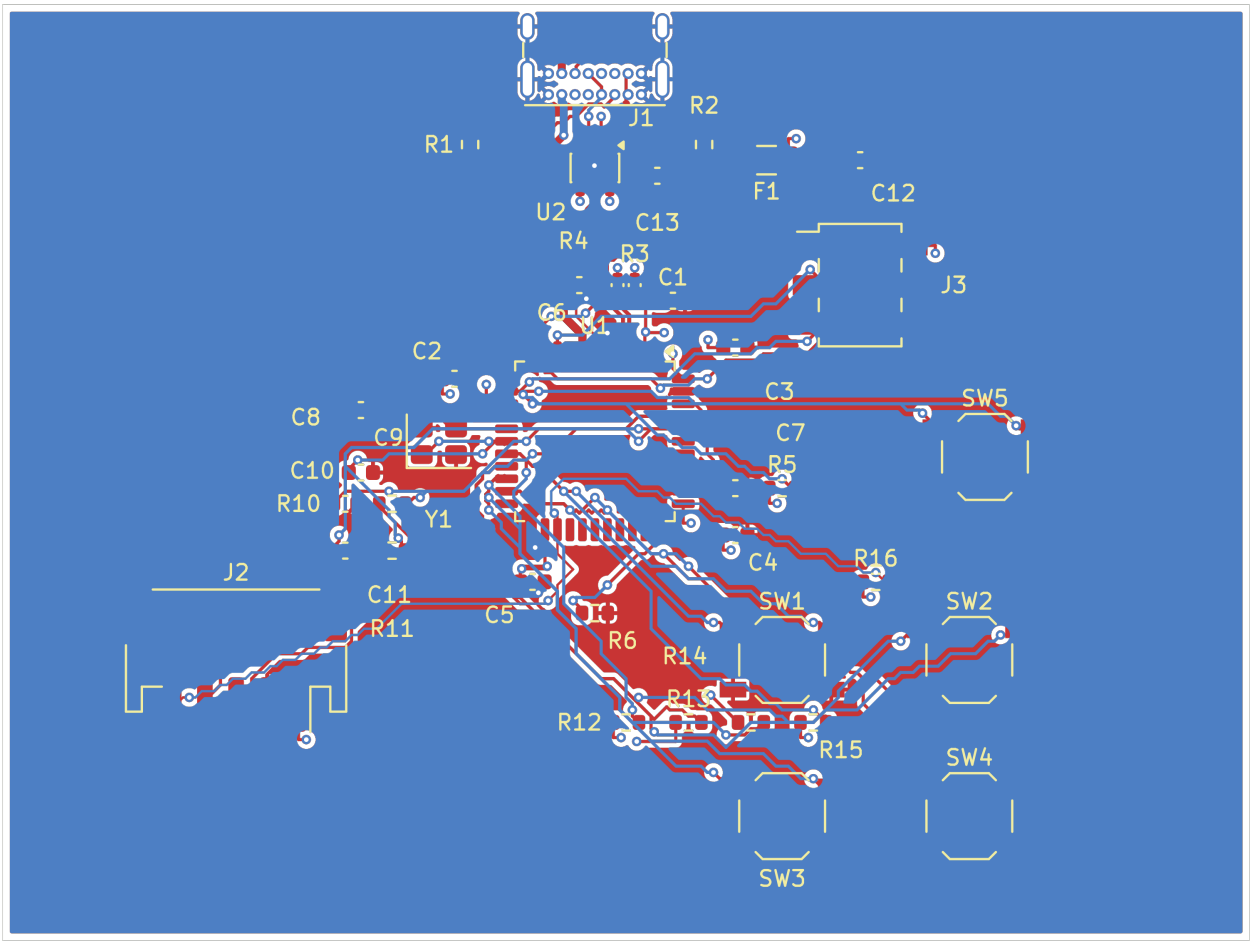


Figure 5: Assembly

Copper Layers

F.Cu

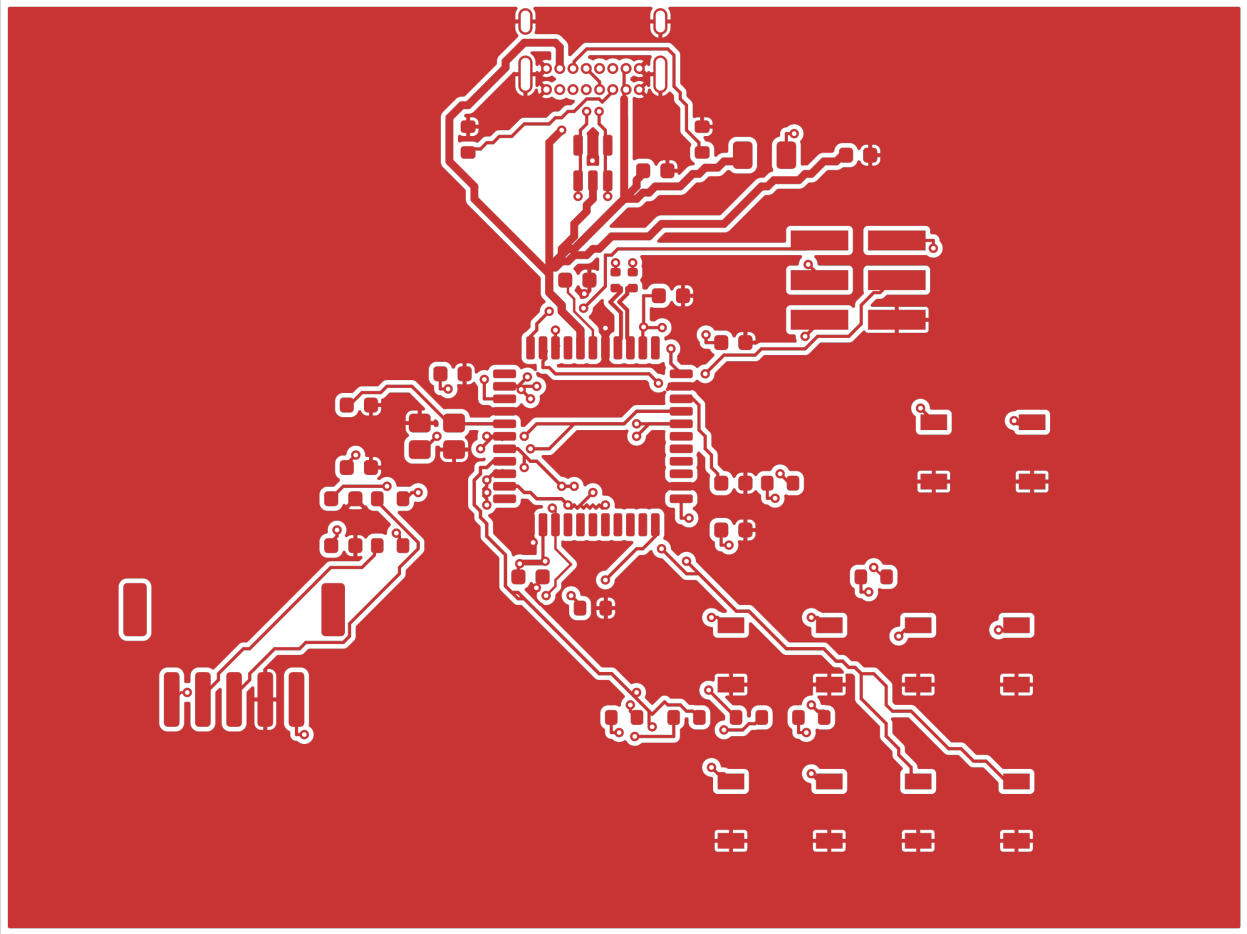


Figure 6: F.Cu

In1.Cu

In2.Cu

B.Cu

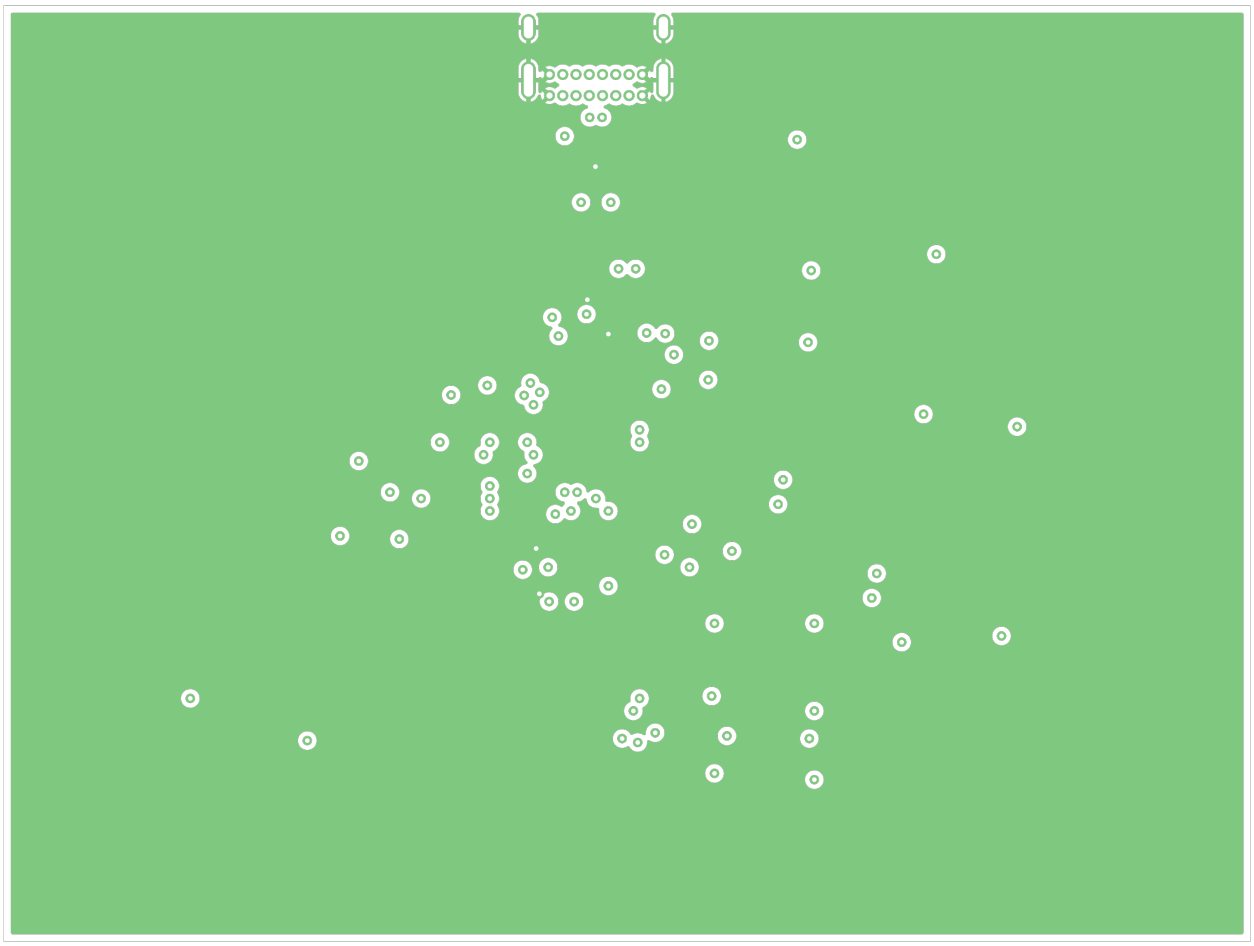


Figure 7: In1.Cu

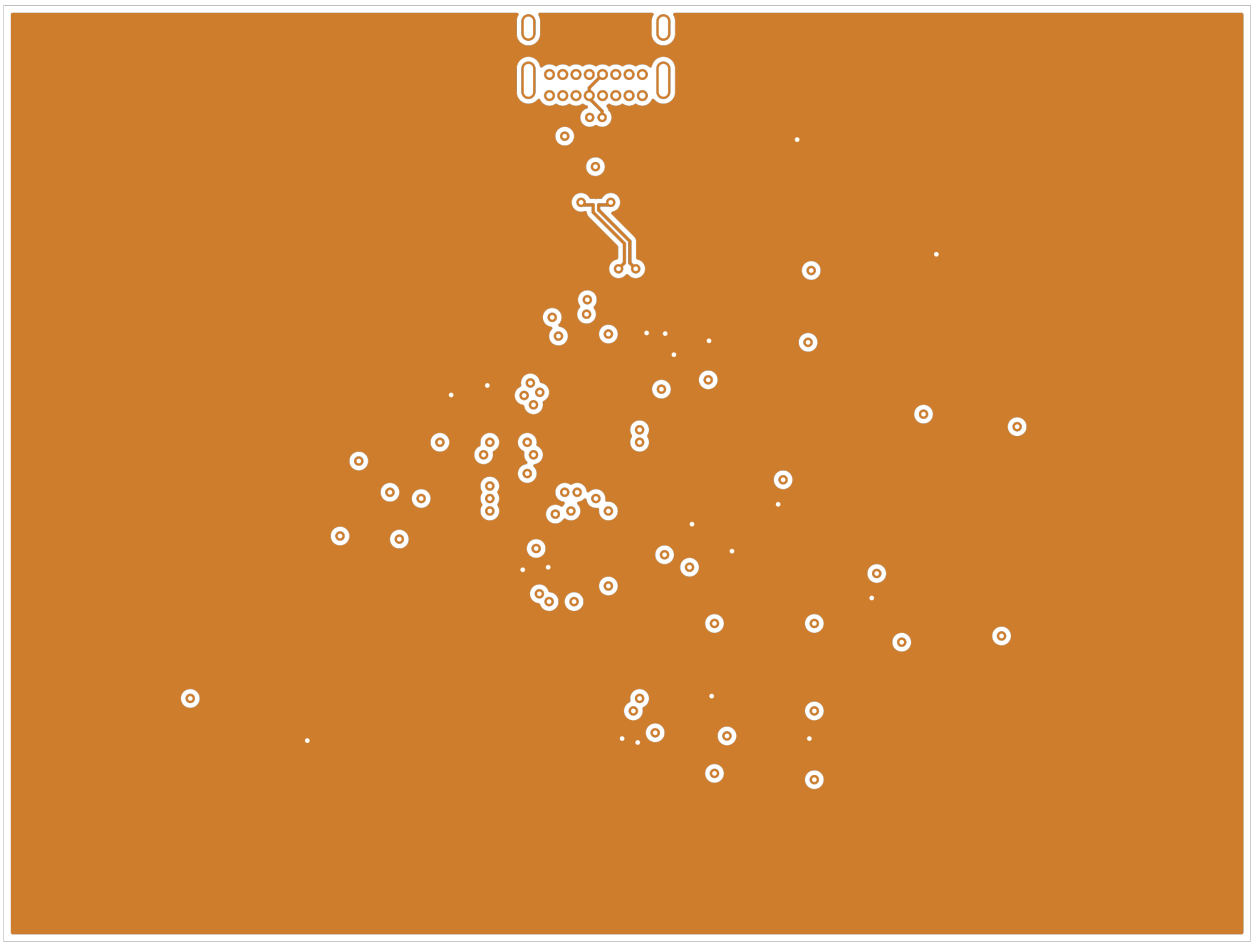


Figure 8: In2.Cu

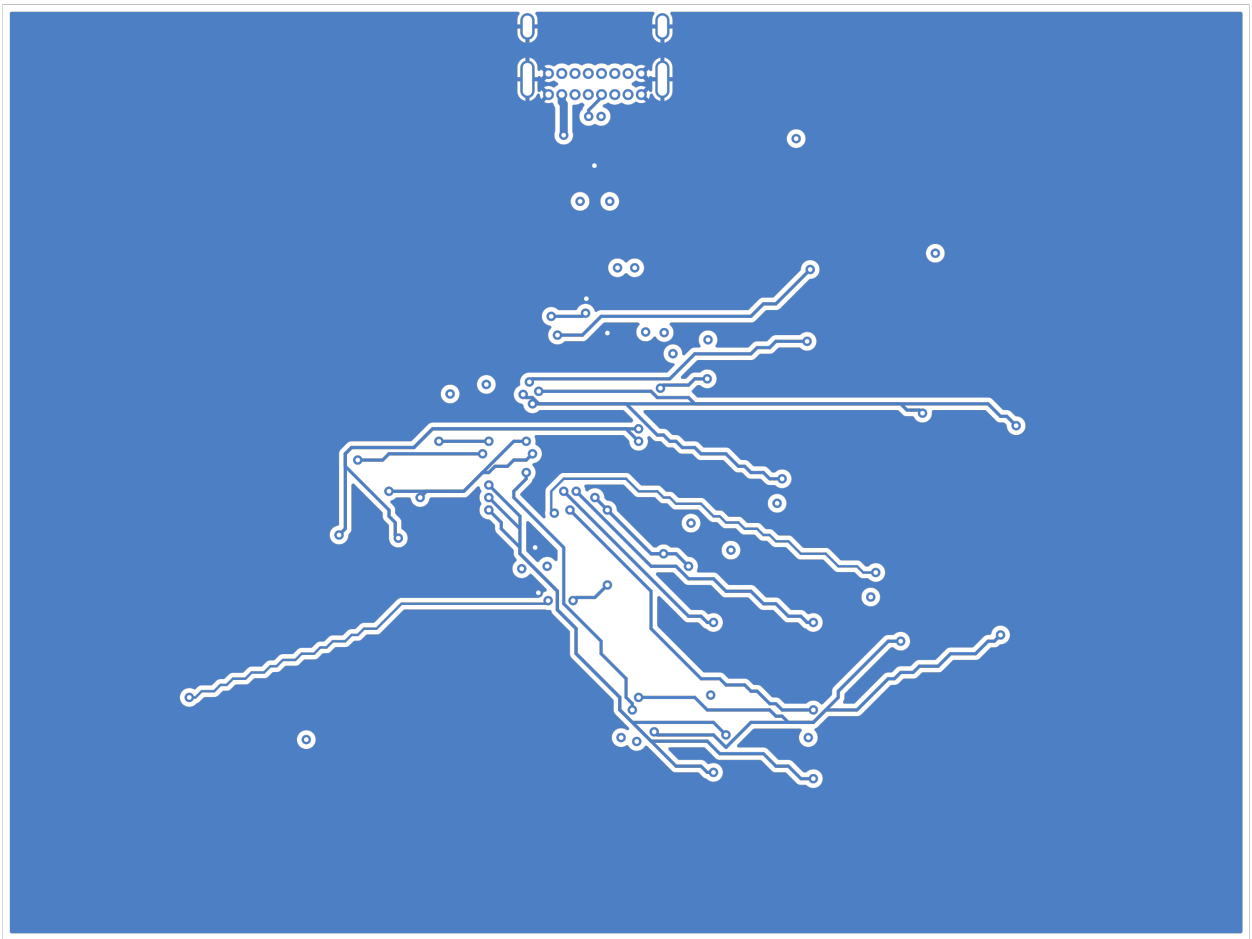


Figure 9: B.Cu

Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0603_1608Metric	7	C1, C2, C3, C4, C5, C7, C13	GRM188R71C104KA01D	C14663
10nF	C_0603_1608Metric	2	C10, C11	GRM188R71H103KA01D	C57112
15pF	C_0603_1608Metric	2	C8, C9	GRM1885C1H150JA01D	C1653
1uF	C_0603_1608Metric	1	C6	GRM188R71A105KA61D	C28323
4.7uF	C_0603_1608Metric	1	C12	GRM188R61A475KE15D	C19666
350mA	Fuse_1206_3216Metric	1	F1	1206L035/16YR	C346817
AVR ISP	Samtec_TSM-103-01-T-DV	1	J3	TSM-103-01-T-DV-P-TR	C21285200
Joystick 5V X Y SW	JST_PH_S5B-PH-SM4-TB_1x05-1MP_P2.00mm_Horizontal	1	J2	S5B-PH-SM4-TB(LF)(SN)	C265104
USB4085-GF-A	USB_C_Receptacle_GCT_USB4085	1	J1	USB4085-GF-A	C7095263
10k	R_0603_1608Metric	7	R5, R6, R12, R13, R14, R15, R16	RC0603FR-0710KL	C98220
1k	R_0603_1608Metric	2	R10, R11	RC0603FR-071KL	C22548
22	R_0402_1005Metric	2	R3, R4	RC0402FR-0722RL	C114765
5.1k	R_0603_1608Metric	2	R1, R2	RC0603FR-075K1L	C105580
TL3342F160QG	SW_SPST_TL3342	5	SW1, SW2, SW3, SW4, SW5	TL3342F160QG	C2886898
ATMEGA32U4-AU	TQFP-44_10x10mm_P0.8mm	1	U1	ATMEGA32U4-AU	C44854
USBLC6-2SC6	SOT-23-6	1	U2	USBLC6-2SC6	C7519
8MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm	1	Y1	ECS-80-12-33-JGN-TR	C2442665

DRC Status (`--mfr jlcpcb-tier1` rules)

Metric	Count
Errors	0
Errors: kct (`--mfr jlcpcb-tier1` rules)	0
Errors: kicad-cli (geometric)	0
Warnings	14
Blocking	0

Status: PASS

Violations by Type

Violation Type	Count
hole_to_hole_clearance	14
diffpair_routing_continuity	1

Manufacturing Readiness

Verdict: WARNING

Action Items

- [**OPTIONAL**] Refill zones to verify 2 zone-connected nets, then re-run kct audit
- [**OPTIONAL**] Review 14 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	100.0% (25/25)
Overall Completion	100.0%
Complete Nets	27 / 27
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND
- VCC

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~2.96 USD
Components (estimated)	~2.11 USD
Assembly (estimated)	~2.16 USD
Total (estimated)	**~7.23 USD**
Batch Quantity	5
Batch Total (estimated)	~36.14 USD